

The background of the entire image is a vibrant red. Overlaid on this background is a large, circular pattern composed of numerous small, light-red dots. These dots are arranged in concentric circles, with the density of the dots increasing towards the center, creating a sense of depth and movement. The dots are slightly offset from each other, giving the pattern a dynamic, almost vibrating appearance. In the center of this circular pattern, the HUST logo is prominently displayed in white.

HUST

ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.

The background is a solid red color. Overlaid on this is a large, circular pattern of small white dots. The dots are arranged in concentric circles, with the density of the dots increasing towards the center, creating a halftone or mesh-like effect. The pattern is centered on the slide.

Design of a Mobile IoT Device for Real-time CH₄ Monitoring



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Presentation of the project

- Goals and targets
- Main checkpoints
- Project progress prevision, gantt diagram, ...



Presentation of the project

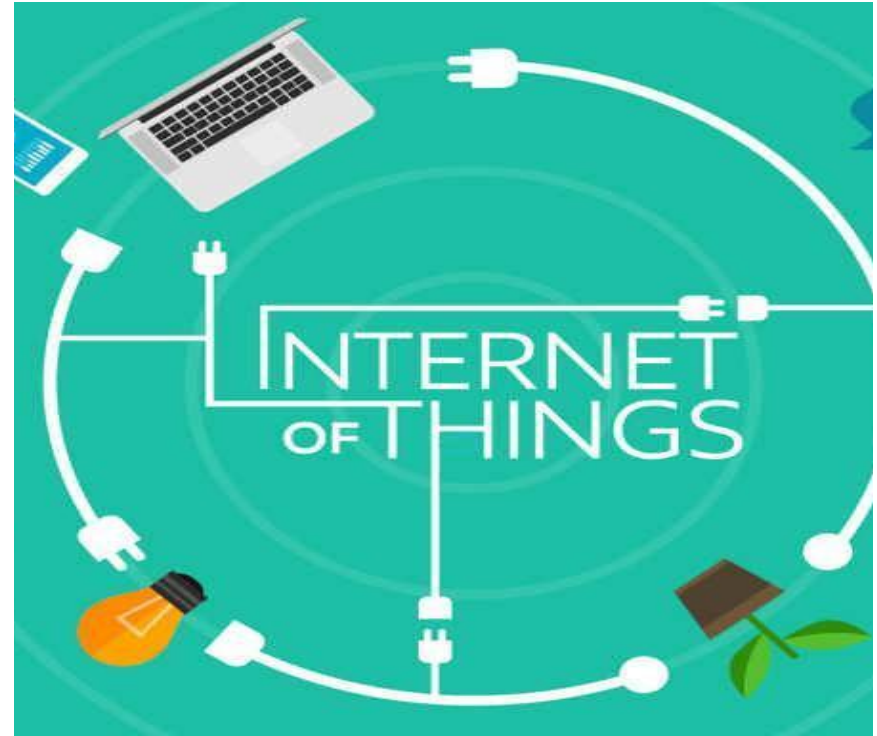
Design and development a mobile IoT device for real-time monitoring of methane gas levels

The device should be able to :

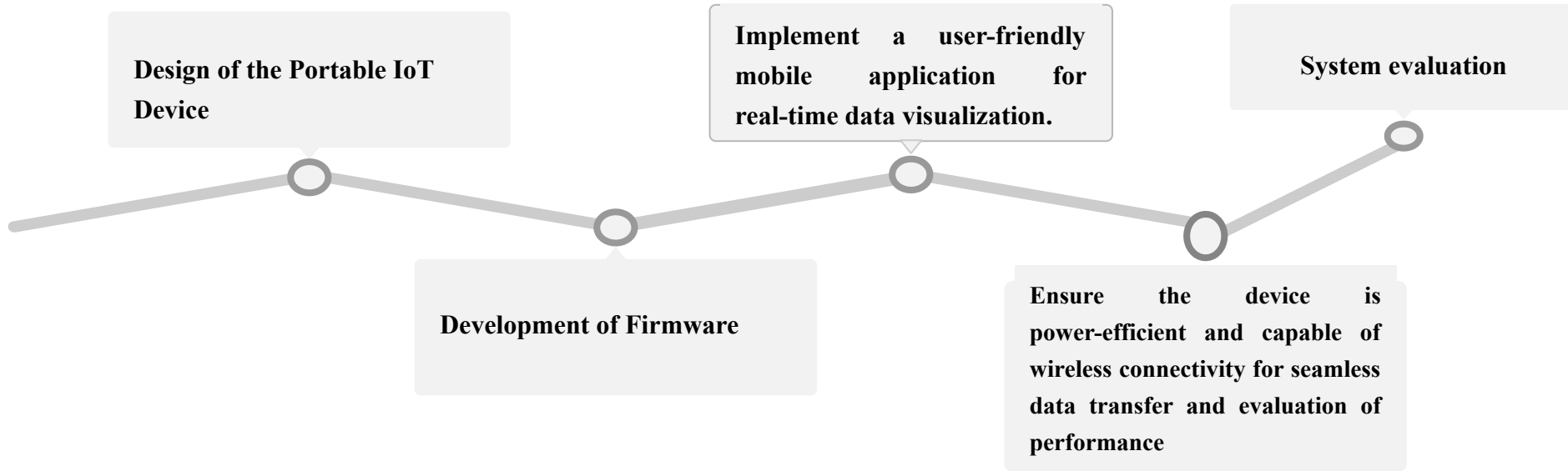
- **measure CH₄ concentrations**
- **transmit data for real-time monitoring and analysis**

This project targets :

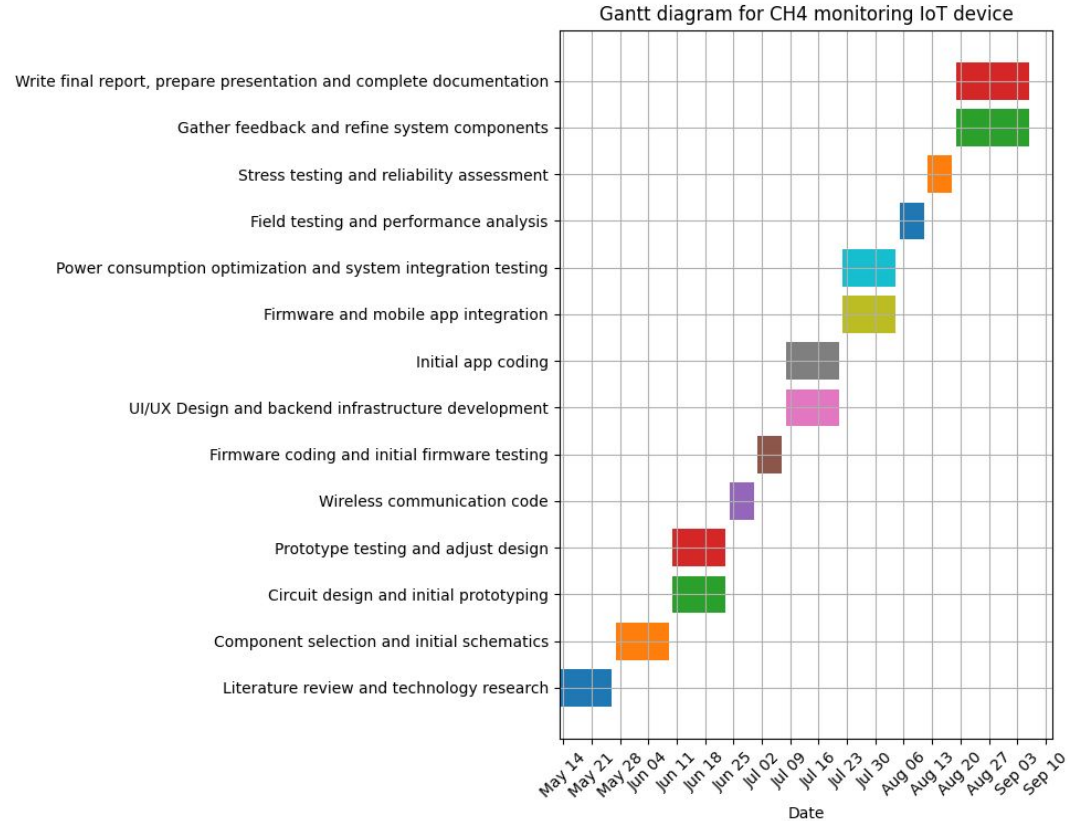
- **agriculture applications**



Presentation of the project



Project Prevision



Gantt diagram for CH4 monitoring IoT device

State of the art

- What is CH₄?
- Why focus on CH₄ instead of other gases?
- How can methane levels be measured?
- What level of precision is required in agriculture?
- What sensors are currently available on the market or being developed by researchers?

Introduction to Methane (CH₄)

Methane Sources:

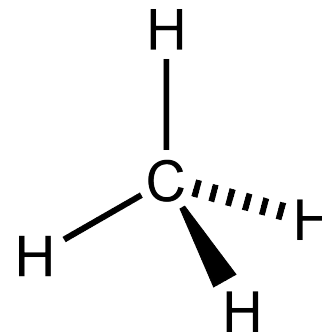
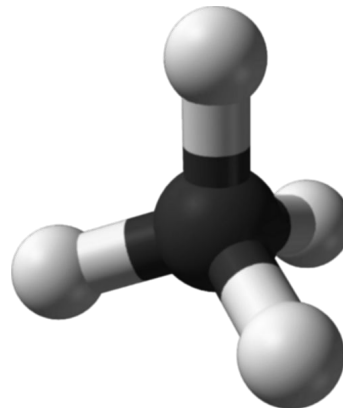
- Naturally present in soil, water, and the atmosphere
- Released by human activities (e.g., agriculture, rice paddies)
- Constitutes 60% of natural gas fossil deposits
- Produced by forced anaerobic fermentation

Uses:

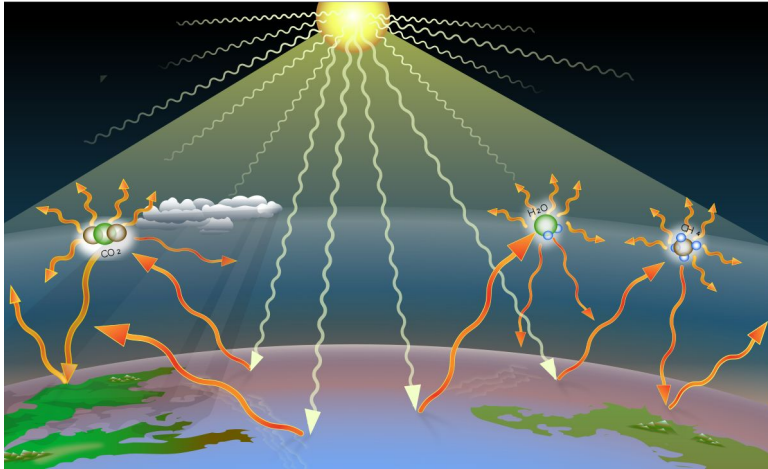
- Electricity, heating, automotive fuel

Hazards:

- Highly flammable and explosive



Environmental Impact of Methane



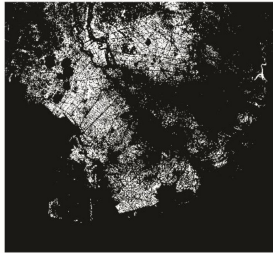
Greenhouse Gas Contribution:

- Second most important greenhouse gas after water vapor
- Absorbs infrared radiation, contributing to global warming

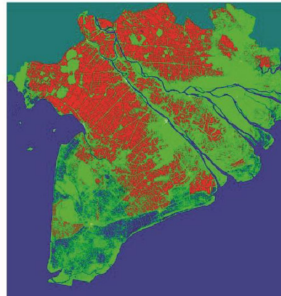
Sources of Emissions:

- Anaerobic fermentation in wetlands, rice paddies, hydroelectric dams
- Fossil fuels

Methane Detection



(a)



(b)

Detection Methods:

- Satellite or on-site sensors

Objective:

- Reducing methane emissions is more cost-effective and efficient for mitigating global warming compared to reducing CO₂

Mitigation Strategies and Gas Sensor Functionality

Strategies:

- Sensors + burning or distillation purification of CH₄ emissions
=> reduction of CO₂ emission
- Modify cattle diets
- Rice cultivation: reduce fertilizer use, combat water turbidity, and reduce methanogenic sediment production



Types of Gas Sensors:

- Electrochemical Sensors: Detect CO and SO₂
- Semiconductor Sensors: Sensitive to a wide range of gases
- Infrared Sensors: Detect CH₄ and CO₂

Market-Available Sensors

MQ4 Sensor:

- Detection range: 200-10000 ppm CH₄

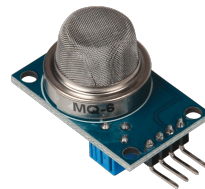
MQ6 Sensor:

- Detection range: 300-10000 ppm (Propane and Methane)

MQ Sensor family :

- Semiconductor sensor
- Long range of sensitivity
- Low price
- Low consumption
- Unrestricted use condition (high range of temperatures resistance, high resistance from humidity, ...)

Tao	Using Tem	-10°C -50°C
Tas	Storage Tem	-20°C -70°C
R _H	Related humidity	less than 95%Rh



Detecting concentration
scope:
200-10000ppm
CH₄ , natural gas

Extract of MQ4 datasheet

Market-Available Sensors

Component	Manufacturer / Distributor	Pros / Cons	Sensitivity - Accuracy	Detection range	Price	Article link (Hyperlinks)
MQ SEN0565	DFRobot / Mouser Electronics	Need often calibration, Need a known sample of CH4	Low	1-10000ppm	\$4.38	Mouser
DFRobot Gravity: I2C Digital Gas Sensor (CH4)	DFRobot / DFRobot	Need often calibration, Need a known sample of CH4	Low	200-10000ppm	\$7	DFRobot
SGX Sensortech MICS-4514	Amphenol SGX Sensortech / Mouser Electronics	Datasheet with no info	No info	>1000ppm	\$11.5	Mouser
Adafruit MIC5524	Adafruit / Adafruit	Can't tell you which gas it has detected	Low	~ 1,000++ ppm	\$15	Adafruit
CH-A3 Combustible Gas Pellistor	AlphaSense / GasLab	High quality, low power	Sensitivity (mV / % methane): 15 to 22 = 0.00015 mV/ppm	0 to 100% LEL methane	\$159	GasLab
Winsen MH-440D	Winsen/Winsen	Expensive	0.05% VOL	0~100%vol	\$235	Winsen Sensor
MUL13442-Safety-CH4-5%vol	elichens / DigKey	Expensive Auto-calibration Ultra Low Power (Peak current < 4.0 mA)	Accuracy ±0.1% vol. Or ±5% of reading	0...5% vol.	\$273.00	Digkey
MPS Methane Gas Sensor	NeyadaNano / CO2meter	Factory calibrated Detection of gas type Low Power	Resolution 1 ppm	500 - 1,500ppm	\$608.85	CO2meter

==>

A lot more on the market :

- prices ranges very wide
- documentation often inexistant or incomplete
- high delivery delay

Key Steps for Implementation

Embedded Board:

- Find a suitable methane sensor
- Use a microprocessor to acquire data
- Transmit data to the cloud, stored on platforms like ThingSpeak



On-site board design

- On-site Board Requirements
- Power Supply and Autonomy
- Component choices
- Block diagrams of concepts and schematic



On-site Board Requirements

Components:

- Methane sensor
- Communication module
- Battery and autonomous battery charger
- Microcontroller

Power Supply and Autonomy

Power Options:

- Battery-powered system for portability
- Solar-powered recharging system using an on-board solar cell

Environmental Constraints:

- Temperature range: 0°C to 50°C
- Humidity resistance: Up to 95%

Energy Autonomy:

- Renewable energies: solar and wind power

Communication module

Several devices are possible:

- a WiFi device, such as ESP8266 or ESP32
- a low-energy Bluetooth device, such as Nordic nRF52
- a LoRa device, such as RFM95
- a cellular device => self-sufficient



not sufficient on their own
require a gateway



TOP

- ~~Highly optional~~ and protocol access

NB IoT : (Narrowband Internet of Things)

- wireless communication technology
- low power wide area networks
- fights against disparition of 2G and 3G networks and high prices of 4G module
- designed for low power consumption with long distance communication
- licensed spectrum = reliability and security



Microcontroller

Key points to decide :

power consumption

portability

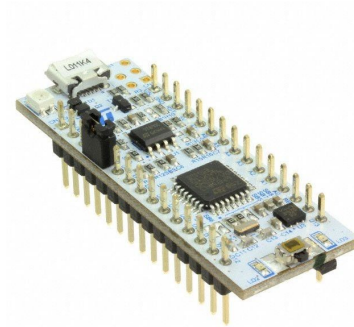
resistance to environmental conditions

board autonomy

Two categories fit :

STM32L0 and **STM32L4**

=> **Ultra-Low Power microcontrollers**



Nucleo-L011K4 with embedded STM32L011K4

Power supply

Battery-powered system to ensure portability

System should be autonomous

=> solar-powered charging module with on-board solar cell

The STM32 Nucleo-32 board and its shield boards can be powered in three different ways from an external power supply, depending on the voltage used. The three power sources are summarized in the [Table 5](#).

Table 5. External power sources

Input power name	Connector pin	Voltage range	Max current	Limitation
VIN	CN4 pin 1	7 V to 12 V	800 mA	From 7 V to 12 V only and input current capability is linked to input voltage: 800 mA input current when VIN=7 V 450 mA input current when 7 V < VIN < 9 V 300 mA input current when 10 V > VIN > 9 V less than 300 mA input current when VIN > 10 V
+5 V	CN4 pin 4	4.75 V to 5.25 V	500 mA	ST-LINK not powered
+3V3	CN4 pin 14	3 V to 3.6 V	-	ST-LINK not powered and SB14 and SB9 must be off.

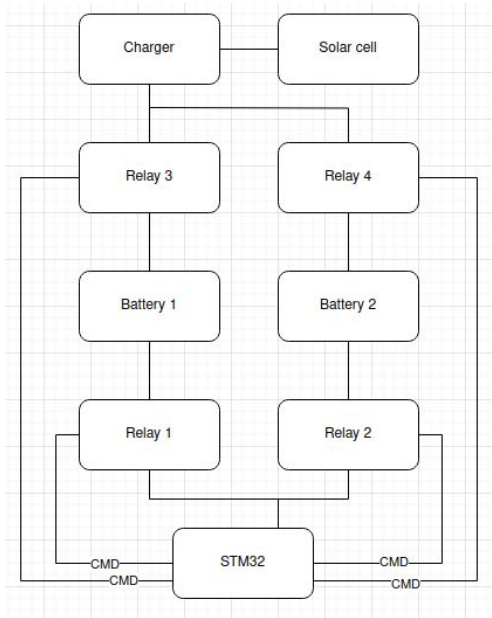


Battery Li-Ion 3.7V 2600mAh

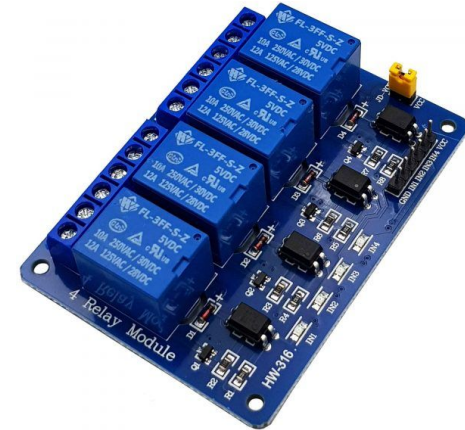
Extract of Nucleo-32 datasheet

Power supply

Autonomous system => two-batteries relay module



Theoric schematic
of charging and supply method



Module 4 Relay 5VDC

Charger and solar cell



**Solar Panel + Battery Charger
STEVAL-ISV012V1**

Includes multiple features :

- protect the battery against overcharging,
- deep discharge, and other potential problems

Composed by :

- SPV1040 Solar Boost Converter
 - L6924D Linear Charger
- => power MOSFET = charging current switch,
reverse blocking diode = no back charge,
battery thermal control

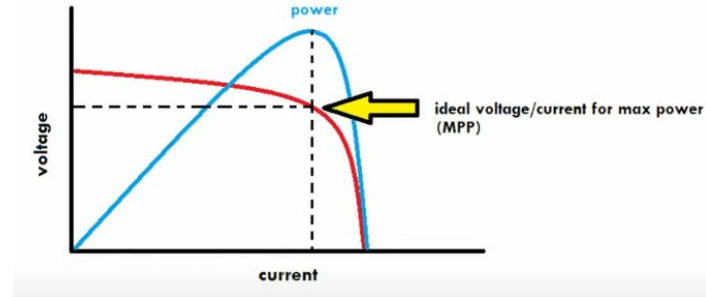
Maximum Power Point Tracking

MPPT :

- essential algorithm used in solar power systems
- used to maximize the efficiency of the energy conversion process
- continuously track MPP and adjust the operating parameters

Maximum Power Point is the specific point on this curve where the product of current (I) and voltage (V) is highest

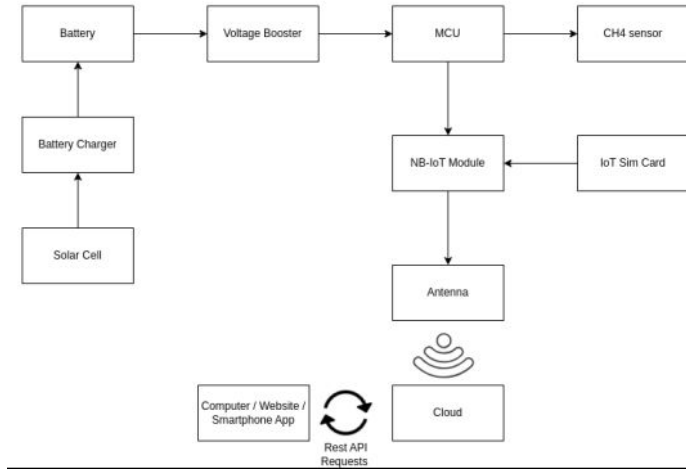
The algorithm : Perturb and Observe (P&O) algorithm



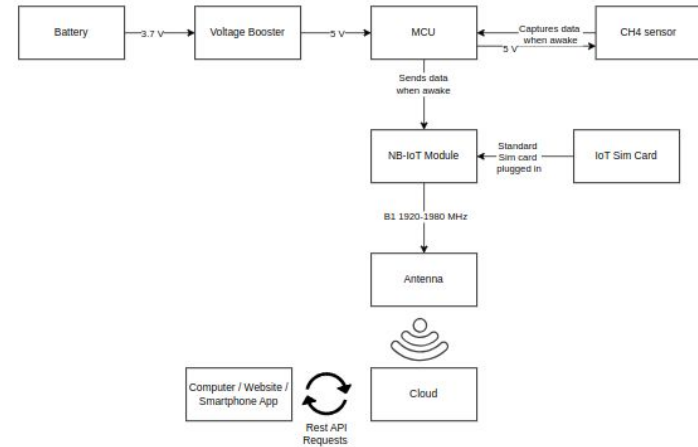
1. **Perturbation:** periodically perturbs (adjusts) the operating voltage or current of the solar panel
2. **Observation:** observes changes
3. **Decision:**
 - If increase in power output, the perturbation is continued in the same direction.
 - If decrease in power output, the perturbation direction is reversed.
4. **Iteration:** This process is repeated continuously

Block diagrams

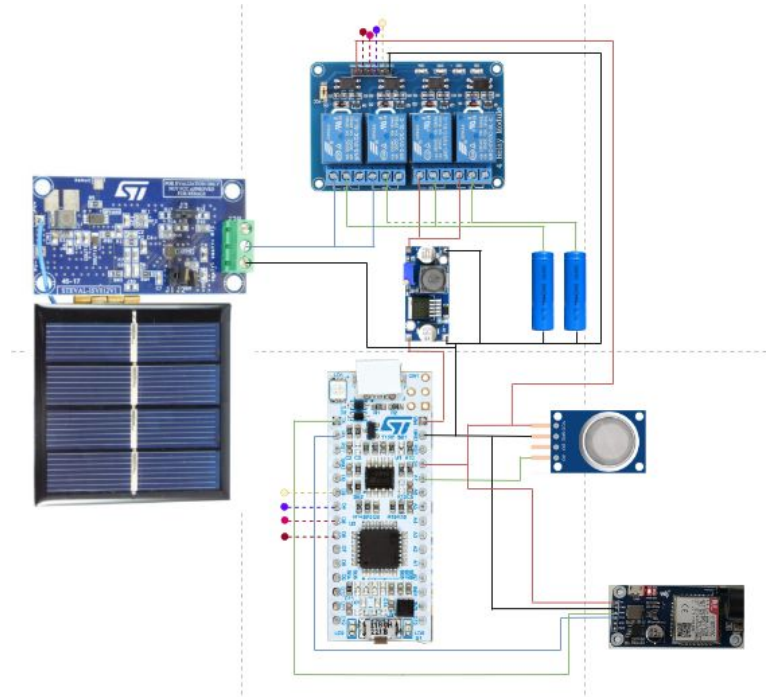
BLOCK DIAGRAM WITH CHARGER



BLOCK DIAGRAM WITHOUT CHARGER



Final schematic of the design



Recap of the design :

- Low Power MCU
- NB IoT communication module
- MQ6 methane sensor
- Solar Cell Charge System
- Autonomous power supply



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Implementation of a user-friendly mobile application for real-time data visualization

- Collaboration with VNUA
- Communication Method
- ThingsBoard



Collaboration with VNUA

The project was conducted in collaboration with Vietnam National University of Agriculture

Their requirements were :

- Design of a user friendly human-machine interface displaying real-time on-site measurements
- Measurements must include :
GPS location, Methane level,
Temperature, Humidity



Communication method

Different communication protocols with SIM7080G :

Protocol Name	HTTP	CoAP	MQTT
Minimum Packet Size	32 bytes (request case) Byte 1-14 : request line Byte 15-30 : host header Byte 31-32 : empty line	4 bytes Byte 1: 2 bits for version, 2 bits for type, and 4 bits for token length. Byte 2 : 8 bits for the code Bytes 3 and 4 : 16 bits for the message ID.	2 bytes Byte 1 : Control packet type and flags Byte 2 : Remaining length
Model	Full-duplex, Bidirectional	Request/Response	Publish/Subscribe
Transport Mechanism	TCP	UDP	TCP, UDP
Messaging Mode	Asynchronous	Synchronous	Asynchronous
Pros	Widely used, easy to implement, supports large payloads.	Lightweight, designed for resource-constrained devices, low overhead.	Very low latency, minimal overhead, optimized for small data packets, supports reliable message delivery with QoS (Quality of Service) levels, widely supported with numerous libraries and tools.
Cons	Higher latency, more overhead, less efficient for small data packets, not optimized for frequent data transmission.	Less mature compared to MQTT, fewer libraries and tools available, might have higher latency in certain network conditions.	Requires a broker for message handling, which adds a layer of complexity.

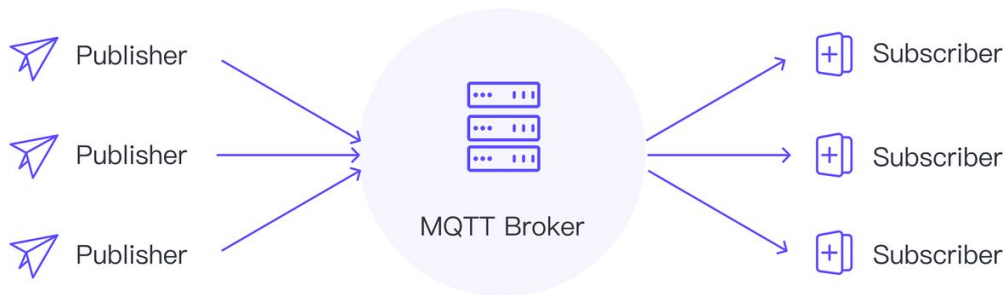
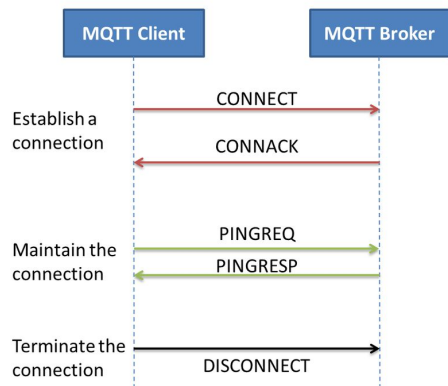
MQTT is the best one for our application:

- low latency and bandwidth
- small packets

Really important for real-time IoT

Message Queuing Telemetry Transport

Principle and connection routine :

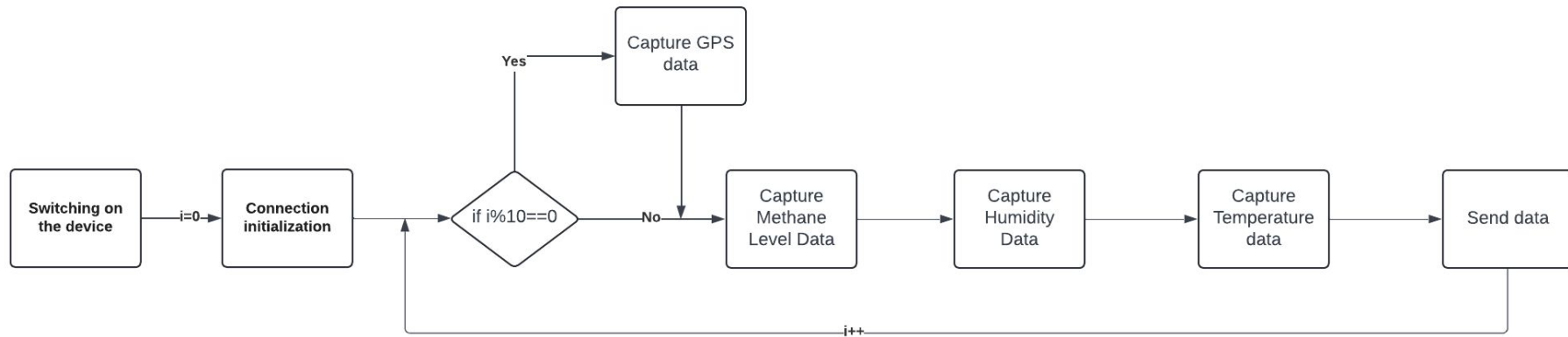


Example of usage :

```
mounsef@mounsef-MacBookAir: ~$ mosquitto_pub -d -q 1 -h mqtt.thingsboard.cloud -p 1883 -t v1/devices/me/telemetry -u "QC2IMzAvHT8pZJhgSUuk" -m "{latitude:20; longitude:105}"
Client (null) sending CONNECT
Client (null) received CONNACK (0)
Client (null) sending PUBLISH (d0, q1, r0, m1, 'v1/devices/me/telemetry', ... (28 bytes))
Client (null) received PUBACK (Mid: 1, RC:0)
Client (null) sending DISCONNECT
```

Communication routine

Postulate : each capture and data sending takes approximately 1s
VNUA requirements : receive a group of data each 5 seconds



But how to elaborate a communication between user interface and STM32 MCU with SIM7080G ?

Attention Commands (or AT Commands)

Set of instructions used to control modems and other communication modules, including IoT devices like the SIM7080G, allowing you to :

- establishing network connections,
- sending data,
- receiving data,
- configuring the device

Examples of commands :

- **AT+SMCONN : connect the broker**
- **AT+SMCONF="URL","mqtt.thingsboard.cloud",1883 : set up MQTT connection**
- **AT+SMPUB="v1/devices/me/telemetry",17,1,1 : publish a data**

Why ThingsBoard ?

- open-source IoT platform for data collection, processing, visualization, and device management
- supports cloud and on-premises deployments,
- combines scalability, fault-tolerance and performances to nullify the chances of getting your IoT data lost.

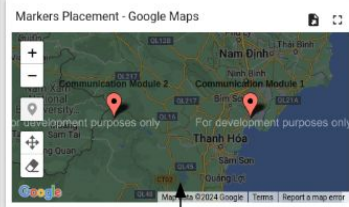
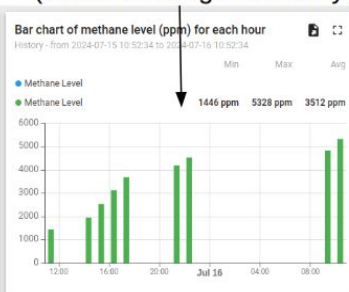
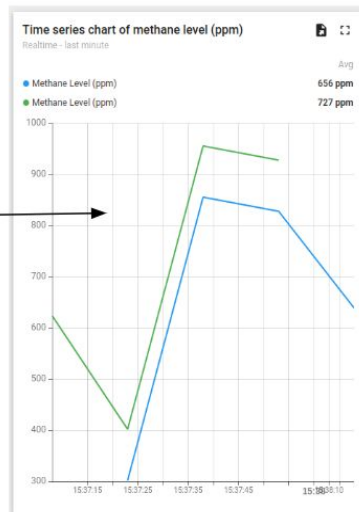
Recap of what interfaces need to contain to fill VNUA requirements :

- real-time methane concentration, temperature and concentrations levels,
- historical data trends,
- alert notifications for threshold breaches

User Interface

Daily mean methane level representation grouping by hour
(can be change manually)

Evolution of methane level in real time in a minute window



Position tracker of multiple devices simultaneously

Alarms

History - last 7 days

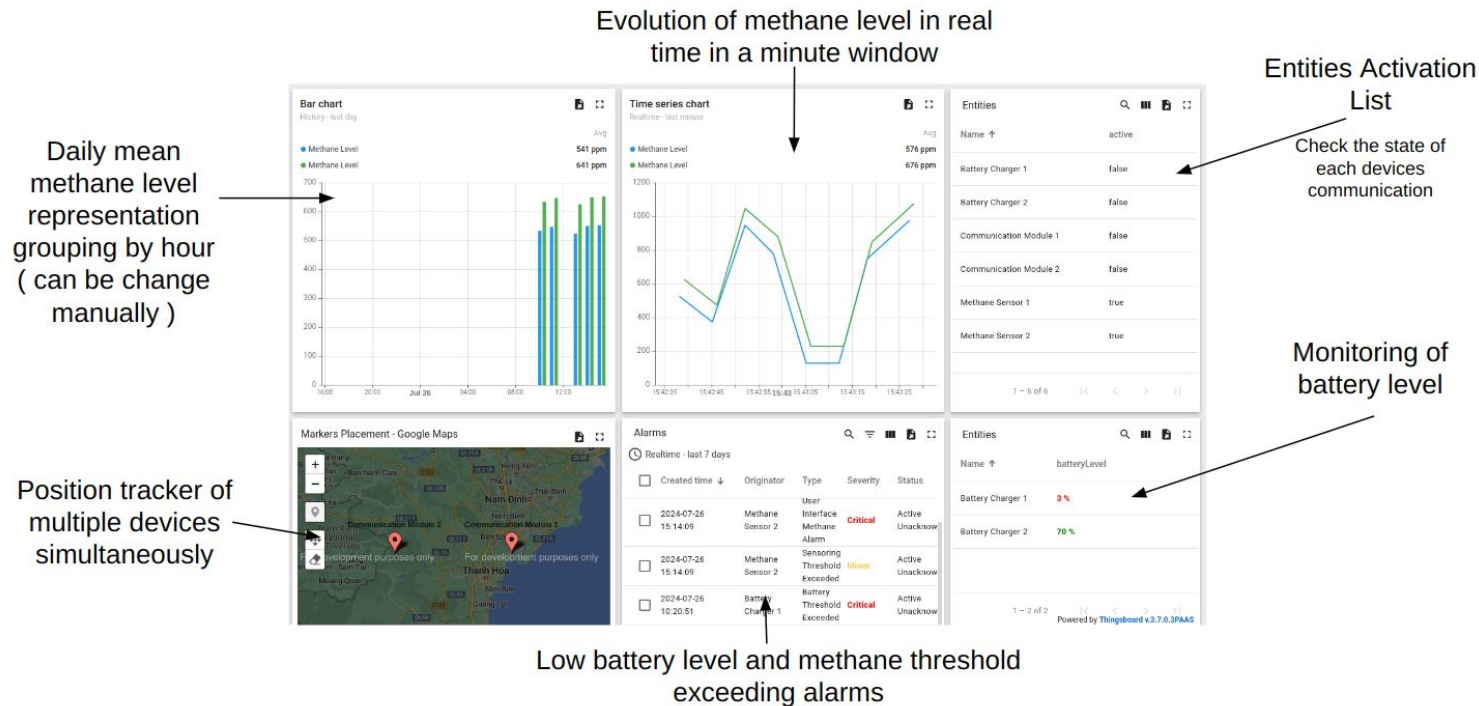
☐	Created time	Originator	Type	Severity	
☐	2024-07-26 15:14:09	Methane Sensor 2	User Interface Methane Alarm	Critical	
☐	2024-07-26 15:14:09	Methane Sensor 2	Sensing Threshold Exceeded	Minor	
☐	2024-07-26 15:13:26	Methane Sensor 1	Sensing Threshold Exceeded	Minor	
☐	2024-07-26 15:13:12	Methane Sensor 1	User Interface Methane Alarm	Critical	
☐	2024-07-26 10:22:55	Methane Sensor 1	User Interface Methane Alarm	Critical	
☐	2024-07-26 10:22:54	Methane Sensor 2	User Interface Methane Alarm	Critical	

2024-07-26 Methane Sensing

1 - 7 of 7

Low battery level and methane threshold exceeding alarms

Administrator Interface





On-site test and results

- Environmental conditions
- Equipment
- Calibration and measurements



Environmental conditions and information about the equipment

Challenges due to the outdoor nature of the fields and the climate of Hanoi :

- hot temperatures,
- high humidity levels
- heavy rainfall
- lack of connectivity

=> Making it difficult to maintain stable measurements and placing additional strain on the electronic components used in the field

The equipment featured in the following pictures and diagrams was provided by another student (Nguyen Bui Thanh An) conducting the same project

Equipment



Core of the system in a protective enclosure
Solar cell and sensors mounted externally
Syringe to calibrate
Isolation box

Measurements



Extracted from ThingsBoards HMI
Even if the methane level seems really stable and low, results seems accurate for the season.

Conclusion

Invaluable experience in terms of the technical knowledge I have gained and the personal growth I have experienced.

- deep understanding of IoT systems, particularly sensor integration, data communication, and the use of platforms like ThingsBoard for data visualization and management.
- hands-on application of theoretical concepts
- underscored the importance of robust design and adaptability

On a personal level I learned :

- the importance of collaboration, communication and adaptability
- insight of value of aligning technical solutions with the specific needs of end-users.



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THANK YOU !



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